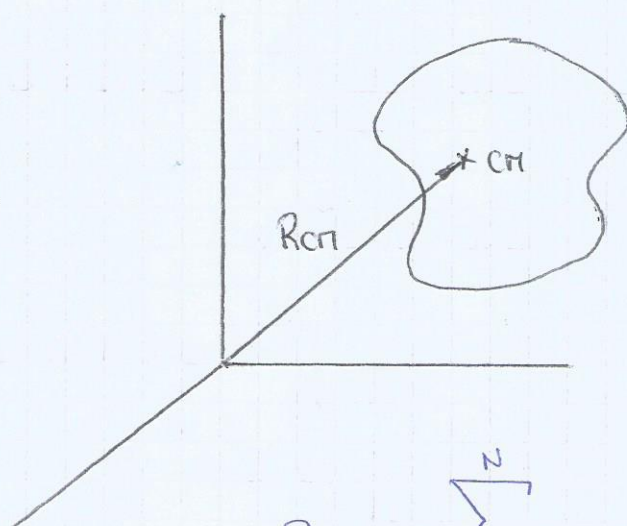


Unidad XII: Dinámica del cuerpo rígido

Dinámica de la traslación:



$$R_{CM} = \frac{\sum_{i=1}^N m_i r_i}{M}$$

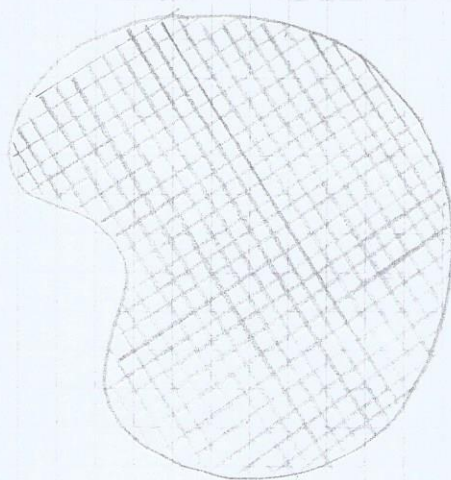
$$F^{ext} = \frac{dP}{dt} = M \cdot \dot{v}_{CM}$$

$$F^{ext} = \frac{dP}{dt} = M \cdot \ddot{R}_{CM}$$

$$L^0_{(CM)} = R_{CM} \times p$$

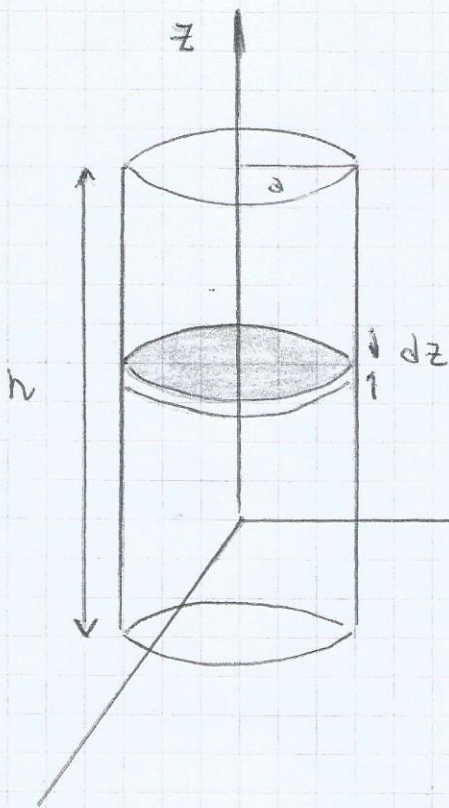
$$\frac{dL_{(CM)}}{dt} = M \cdot \dot{v}_{CM}^{ext} = R_{CM} \times F^{ext}$$

$$\text{con } M = \sum_{i=1}^N m_i$$



$$R_{CM} = \frac{1}{M} \int r \, dm$$

$$R_{CM} = \frac{1}{M} \int_V r \, g(r) \, dv$$



$$g = cz \quad g = g(z)$$

$$R_{cm} = (x_{cm}, y_{cm}, z_{cm})$$

$$x_{cm} = y_{cm} = 0$$

$$z_{cm} = \frac{1}{M} \int_0^h z g(z) \pi a^2 dz$$

$$dV = \pi a^2 dz$$

$$z_{cm} = \frac{1}{M} \int_0^h z cz \pi a^2 dz$$

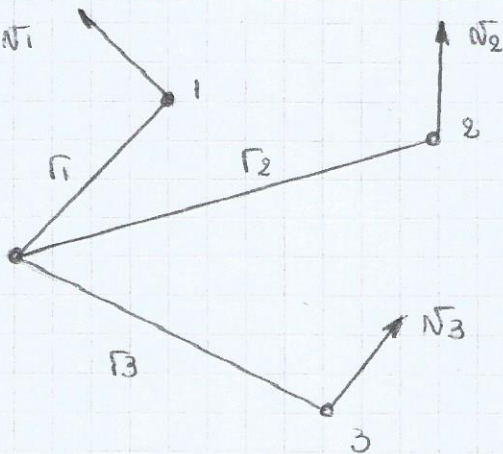
$$z_{cm} = \frac{1}{M} c \pi a^2 \frac{h^3}{3}$$

$$M = \int_V g dV = \int_0^h cz \pi a^2 dz$$

$$= c \pi a^2 \frac{h^2}{2}$$

$$z_{cm} = \frac{2}{3} h$$

Dinámica de la rotación:



$$E_c = \sum_i E_{ci}$$

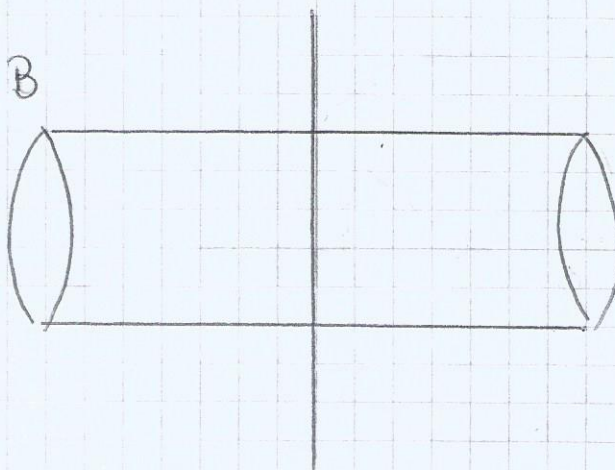
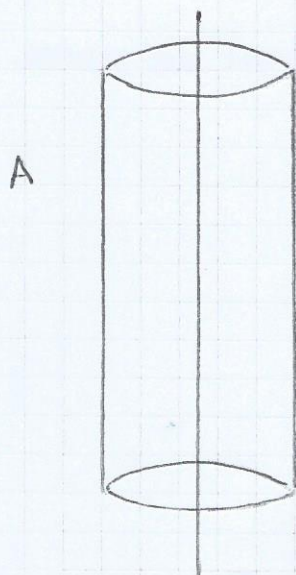
$$\sum_i \frac{1}{2} m_i v_i^2$$

$$v_i = r_i \omega$$

$$I_0 = \sum_i m_i r_i^2$$

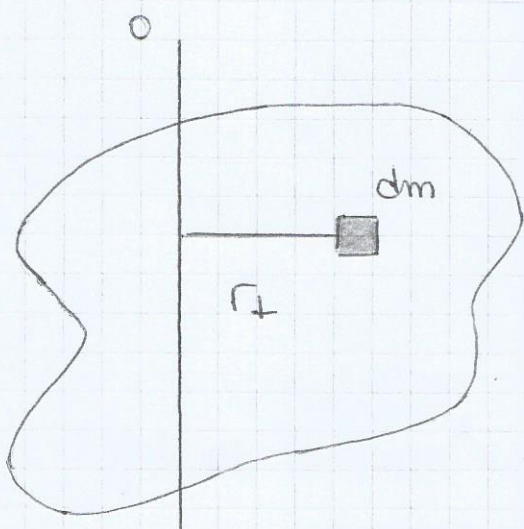
$$E_c = \frac{1}{2} \sum_i m_i \underline{r_i^2} \omega^2$$

Momento de
inercia I_0



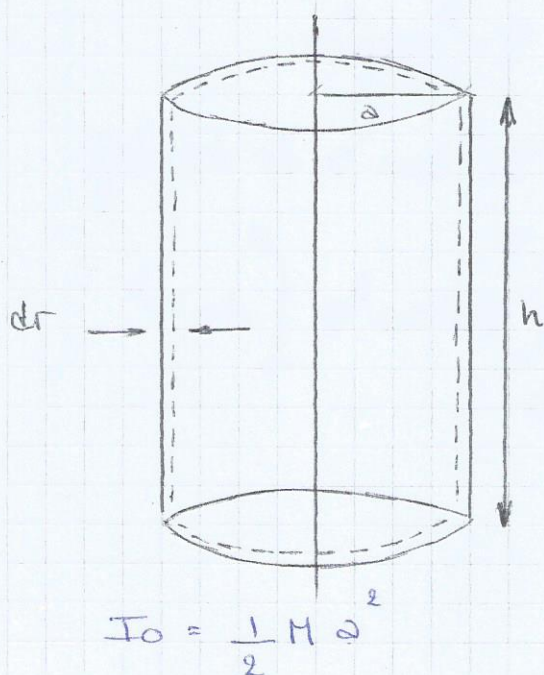
$$I_B > I_A$$

$$E_c(A) < E_c(B)$$



$$I_0 = \int r_{\perp}^2 dm$$

$$I_0 = \int_V r_{\perp}^2 \rho(r) dV$$



$$\rho = \frac{dm}{dV} = \frac{M}{V}$$

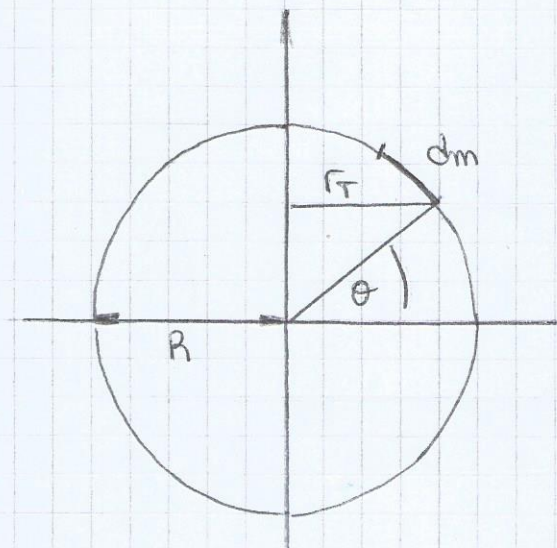
$$dV = 2\pi r h dr$$

$$I_0 = \int_0^a r^2 \rho 2\pi r h dr$$

$$I_0 = 2\pi h \rho \int_0^a r^3 dr = \pi h \rho \frac{a^4}{2}$$

$$M = \int \rho dV = \int_0^a \rho 2\pi r h dr =$$

$$= 2\pi h \rho \frac{a^2}{2}$$



λ densidad lineal de masa

$$dm = \lambda dl$$

$$dl = R d\theta$$

$$r_{\perp} = r \cos \theta$$

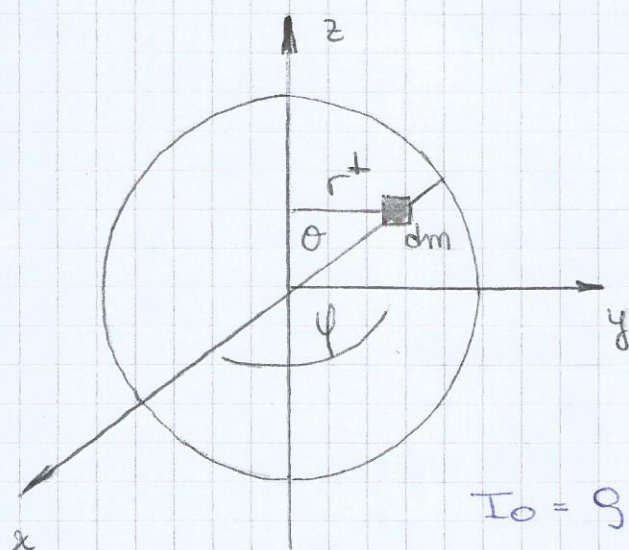
$$I_0 = \int_0^{2\pi} R^2 \cos^2 \theta \lambda R d\theta$$

$$M = \int_0^{2\pi} \lambda R d\theta = 2\lambda\pi R$$

$$I_0 = \lambda R^3 \left(\frac{1}{2} \theta + \frac{1}{4} \sin 2\theta \right) \Big|_0^{2\pi}$$

$$I_0 = \lambda \pi R^3$$

$$I_0 = \frac{M R^2}{2}$$



$$dV = r^2 \sin \theta dr d\theta d\phi$$

$$r_{\perp} = r \sin \theta$$

$$I_0 = \int 9 r^2 \sin^2 \theta$$

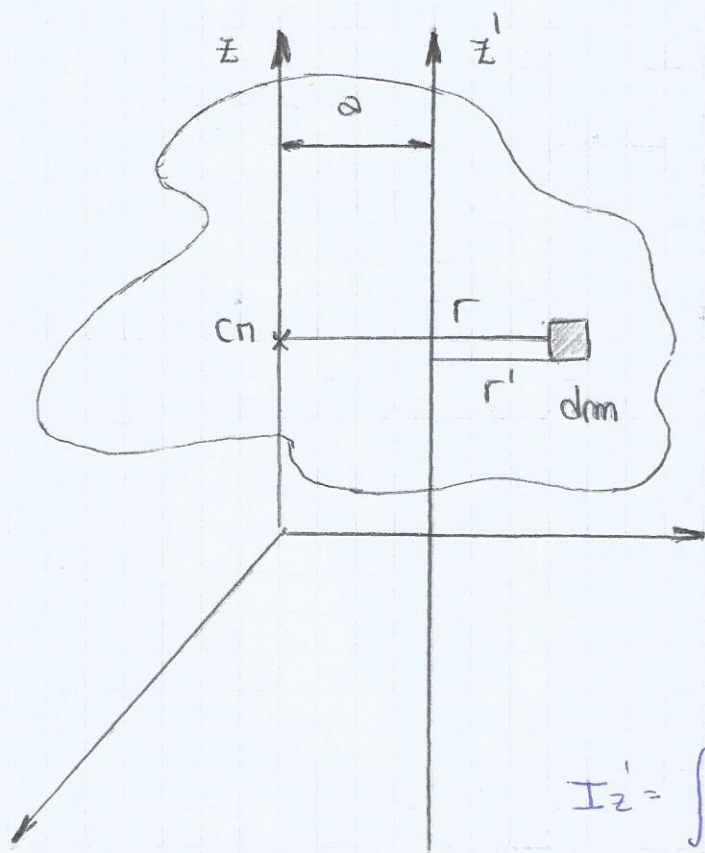
$$I_0 = \int 9 r^4 \sin^4 \theta dr d\theta d\phi$$

$$I_0 = 9 \int_0^{2\pi} d\phi \int_0^{\pi} d\theta \sin^3 \theta \int_0^R dr r^4$$

$$I_0 = \frac{8}{15} 9 \pi R^5$$

$$M = 9 \frac{4}{3} \pi R^3$$

$$I_0 = \frac{2}{5} M R^2$$

Teorema de Steiner:

$$z' \parallel z$$

$$r^2 = x^2 + y^2$$

$$r'^2 = x^2 + (y - a)^2$$

$$I_z = \int_M r^2 dm$$

$$I_z = \int_M (x^2 + y^2) dm$$

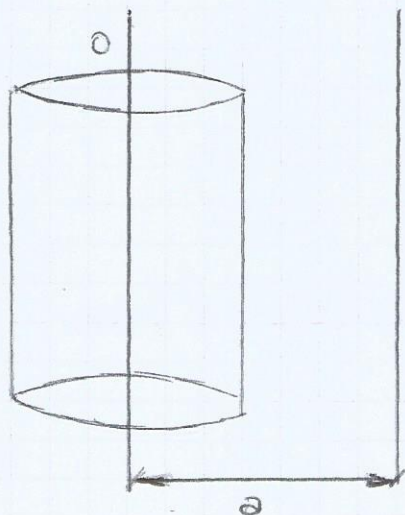
$$I_{z'} = \int_M r'^2 dm$$

$$I_{z'} = \int_M (x^2 + y^2 + a^2 - 2ya) dm$$

$$I_{z'} = \underbrace{\int_M (x^2 + y^2) dm}_{I_z} + \underbrace{a^2 \int_M dm}_{Ma^2} - \underbrace{2a \int_M y dm}_0$$

$$I_{z'} = I_z + Ma^2$$

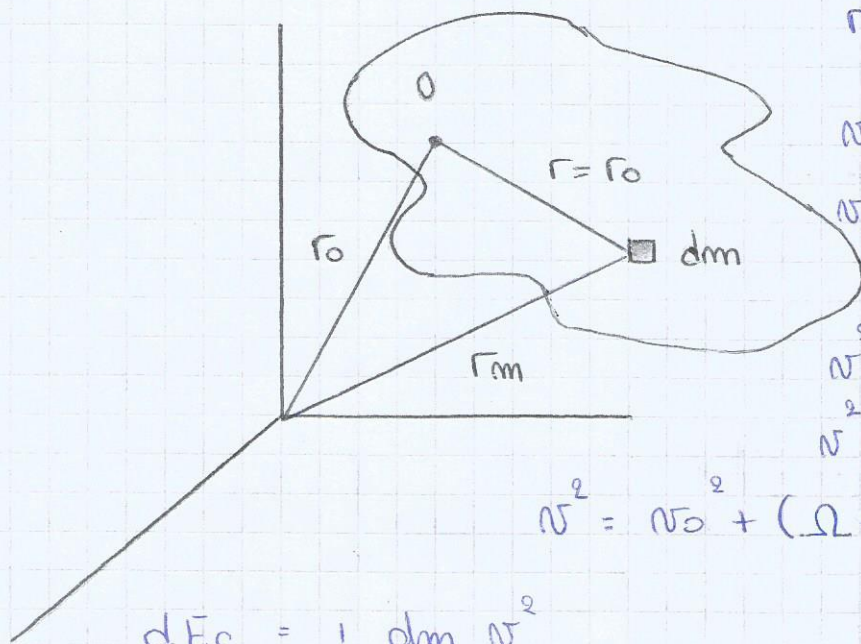
(con a la distancia mínima entre ejes)



$$I_O = \frac{MR^2}{2}$$

$$I_{O'} = \frac{MR^2}{2} + Ma^2$$

Movimiento de rototranslación:



$$\mathbf{r}' = \mathbf{r} - \mathbf{r}_0$$

$$\mathbf{v} = \mathbf{v}_0 + \boldsymbol{\Omega} \times (\mathbf{r} - \mathbf{r}_0)$$

$$\mathbf{v} = \mathbf{v}_0 + \boldsymbol{\Omega} \times \mathbf{r}'$$

$$v^2 = \mathbf{v} \cdot \mathbf{v}$$

$$v^2 = (\mathbf{v}_0 + \boldsymbol{\Omega} \times \mathbf{r}') \cdot (\mathbf{v}_0 + \boldsymbol{\Omega} \times \mathbf{r}')$$

$$v^2 = v_0^2 + (\boldsymbol{\Omega} \times \mathbf{r}')^2 + 2\mathbf{v}_0 \cdot (\boldsymbol{\Omega} \times \mathbf{r}')$$

$$dE_c = \frac{1}{2} dm v^2$$

$$E_c = \int_M \frac{1}{2} dm v^2$$

$$E_c = \underbrace{\int_M \frac{1}{2} v_0^2 dm}_{\text{Energía cinética de traslación}} + \underbrace{\int_M \frac{1}{2} |\boldsymbol{\Omega} \times \mathbf{r}'|^2 dm}_{\text{Energía cinética de rotación}} + \underbrace{\int_M \mathbf{v}_0 \cdot (\boldsymbol{\Omega} \times \mathbf{r}') dm}_{\text{Energía cinética de rototranslación}}$$

Energía cinética
de traslación

Energía cinética
de rotación

Energía cinética
de rototranslación

$$\begin{aligned} E_c(\text{rot}) &= \frac{1}{2} \int_M \Omega^2 r_{\perp}^2 dm \\ &= \frac{1}{2} \Omega^2 \int_M r_{\perp}^2 dm \\ &= \frac{1}{2} \Omega^2 I_0 \end{aligned}$$

$r_{\perp}$ distancia de dm a $O \perp \boldsymbol{\Omega}$

$$E_c = \frac{1}{2} v_0^2 M + \frac{1}{2} \Omega^2 I_0 + \int_M \mathbf{v}_0 \cdot (\boldsymbol{\Omega} \times \mathbf{r}') dm$$

$$\text{Si } O \equiv \text{cm}$$

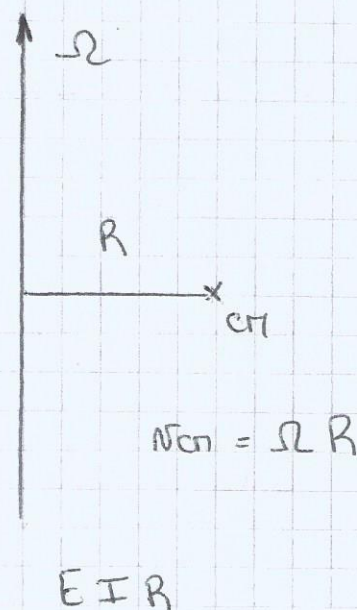
$$E_c(\text{rotot}) = M \text{cm} \int_m \Omega \times r' dm$$

$$E_c(\text{rotot}) = M \text{cm} \cdot \Omega \times \underbrace{\int_m r' dm}_0 \quad (R \text{cm} = 0)$$

$$E_c = \frac{1}{2} M \Omega \text{cm}^2 + \frac{1}{2} I_{\text{cm}} \Omega^2$$

$$\text{Si } O \equiv E.I.R$$

$$E_c = \frac{1}{2} I_0 \Omega^2$$

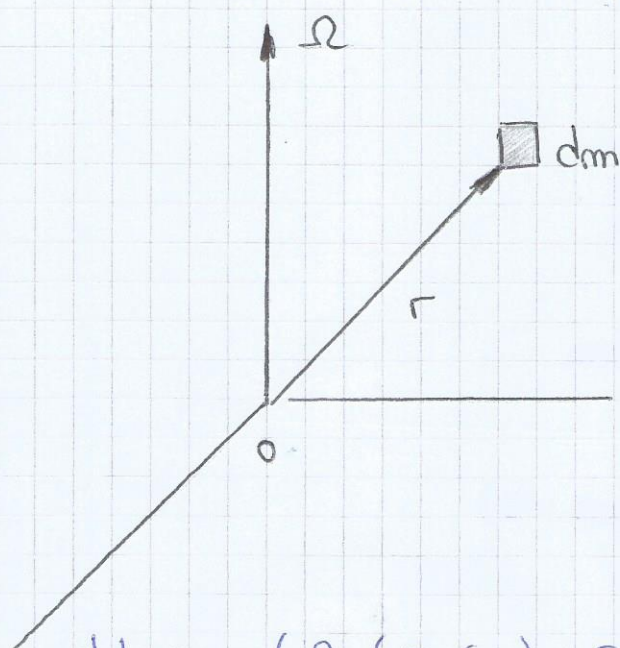


$$E_c = \frac{1}{2} M R^2 \Omega^2 + \frac{1}{2} I_{\text{cm}} \Omega^2$$

$$E_c = \frac{1}{2} \underbrace{(I_{\text{cm}} + M R^2)}_{I_0} \Omega^2$$

Traslación: $\vec{F}^{\text{ext}} = \frac{d\vec{p}}{dt} = m \vec{a}_{\text{cm}}$ Rotación: $\vec{L}_0^{\text{cm}} = \vec{R}_{\text{cm}} \times \vec{p}_{\text{ext}}$
 $\frac{d\vec{L}_0^{\text{cm}}}{dt} = \vec{M}_0$
 $\vec{a}_{\text{cm}} = \ddot{\vec{R}}_{\text{cm}}$

Impulso angular del cuerpo rígido



$$dL_{\Omega} = \mathbf{r} \times \mathbf{v} \, dm$$

$$\mathbf{v} = \boldsymbol{\Omega} \times \mathbf{r}$$

$$dL_{\Omega} = \mathbf{r} \times (\boldsymbol{\Omega} \times \mathbf{r}_{\perp}) \, dm$$

$\mathbf{r}_{\perp}$ componente de $\mathbf{r} \perp \boldsymbol{\Omega}$

$$dL_{\Omega} = (\boldsymbol{\Omega} (\mathbf{r} \cdot \mathbf{r}_{\perp}) - \mathbf{r}_{\perp} (\boldsymbol{\Omega} \cdot \mathbf{r})) \, dm$$

$$dL_{\Omega} = (\boldsymbol{\Omega} r_{\perp}^2 - \mathbf{r}_{\perp} \boldsymbol{\Omega} r_{||}) \, dm$$

$$L_{\Omega} = \int_M (\boldsymbol{\Omega} r_{\perp}^2 - \mathbf{r}_{\perp} \boldsymbol{\Omega} r_{||}) \, dm = \underbrace{\boldsymbol{\Omega} \int_M r_{\perp}^2 \, dm}_{I_{\Omega} \cdot \boldsymbol{\Omega}}$$

$$L_{\Omega} = -\boldsymbol{\Omega} \int_M r_{||} r_{\perp} \, dm$$

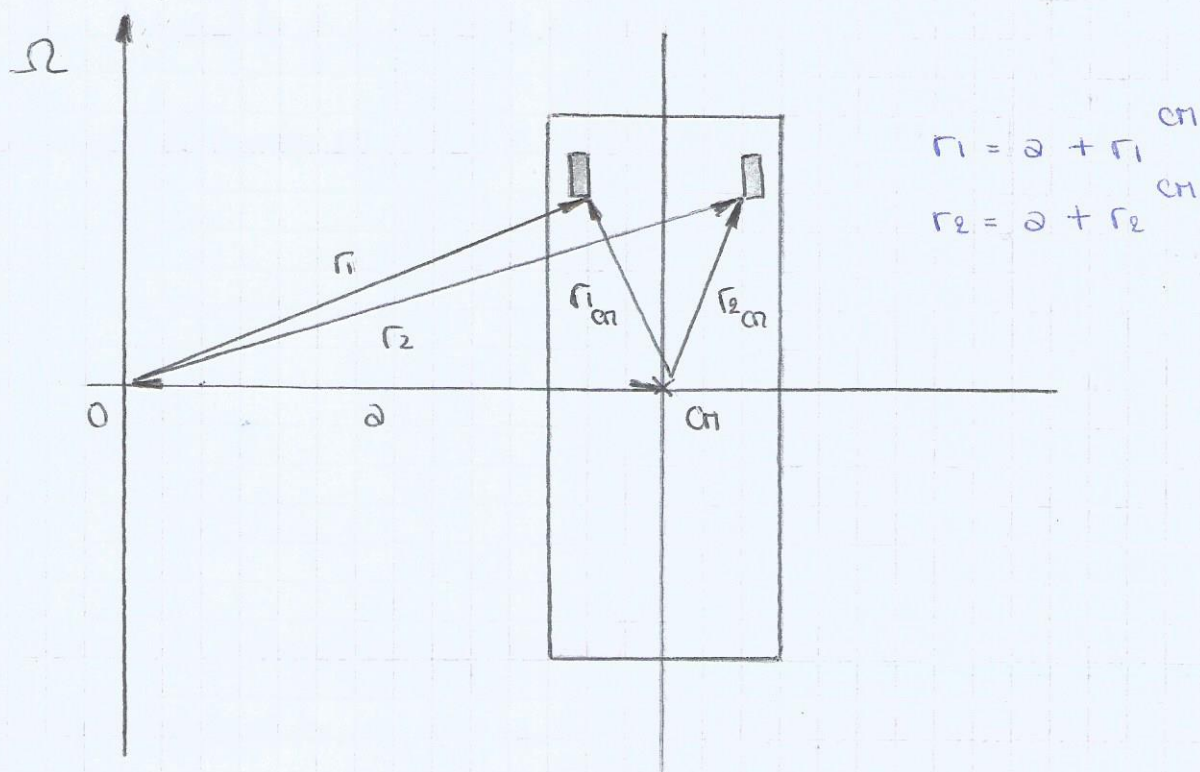
En general, el impulso angular de un cuerpo rígido:

$$L_{\text{cr}} = I_{\Omega} \cdot \boldsymbol{\Omega} + L_{\perp}$$

$$\frac{dL_{\Omega}}{dt} = I_{\Omega} \dot{\boldsymbol{\Omega}} + \dot{L}_{\perp}$$

$$M^{\text{ext}} = I_{\Omega} \dot{\boldsymbol{\Omega}} + \dot{L}_{\perp}$$

$$M^{\text{ext}} = I_{\Omega} \dot{\boldsymbol{\Omega}} + \dot{L}_{\perp}$$



$$r_1 = a + r_{1cm}$$

$$r_2 = a - r_{2cm}$$

$$r_{1\perp} = -r_{2\perp} = r_{\perp}$$

$$r_{1\parallel} = r_{2\parallel} = r_{\parallel}$$

$$r_{1\perp} = a + r_{\perp}$$

$$r_{2\perp} = a - r_{\perp}$$

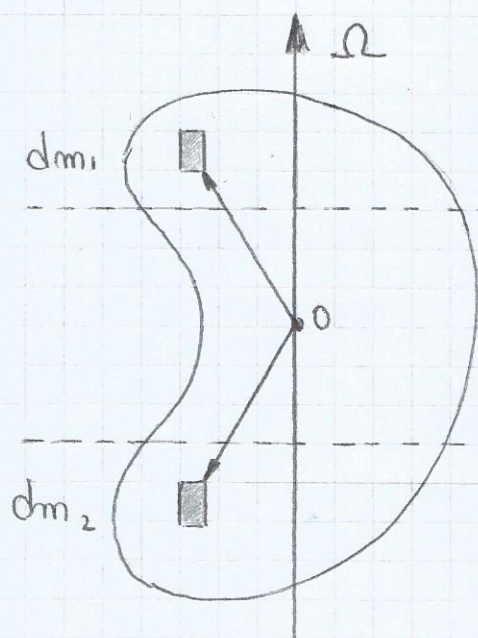
$$dm_1 = dm_2 = dm$$

$$dL_{\perp}^0 = dL_{\perp 1}^0 + dL_{\perp 2}^0$$

$$= \Omega r_{\parallel} (a + r_{\perp}) dm + \Omega r_{\parallel} (a - r_{\perp}) dm$$

$$L_{\perp}^0 = \int_M 2 \Omega r_{\parallel} a dm$$

$$L_{\perp}^0 = 2 a \Omega \int_M r_{\parallel} dm = 0$$



$$dL_{\perp}^0 = dL_{\perp 1}^0 + dL_{\perp 2}^0$$

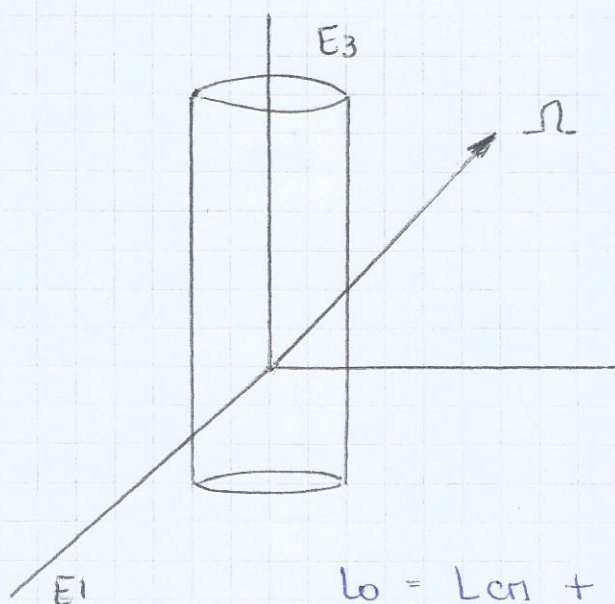
$$dL_{\perp}^0 = \Omega r_{\perp 1} r_{\perp 1} dm + \Omega r_{\perp 2} r_{\perp 2} dm$$

$$dL_{\perp}^0 = 0$$

$$L_{\perp}^0 = 0$$

Ejes principales de inercia: Pasan por el centro de masa (y por lo menos 3 $\perp$ entre si.) $L_{\perp} = 0$, si el cuerpo rígido rota con respecto a algunos de ellos.

$$\Omega \parallel E_p I \Rightarrow L_{\perp} = 0$$



$$\Omega = \Omega_1 \hat{e}_1 + \Omega_2 \hat{e}_2 + \Omega_3 \hat{e}_3$$

$$L_{\Omega_1} = I_1 \Omega_1 \hat{e}_1$$

$$L_{\Omega_2} = I_2 \Omega_2 \hat{e}_2$$

$$L_{\Omega_3} = I_3 \Omega_3 \hat{e}_3$$

$$L_0 = L_{cm} + R^{cm} \times p$$

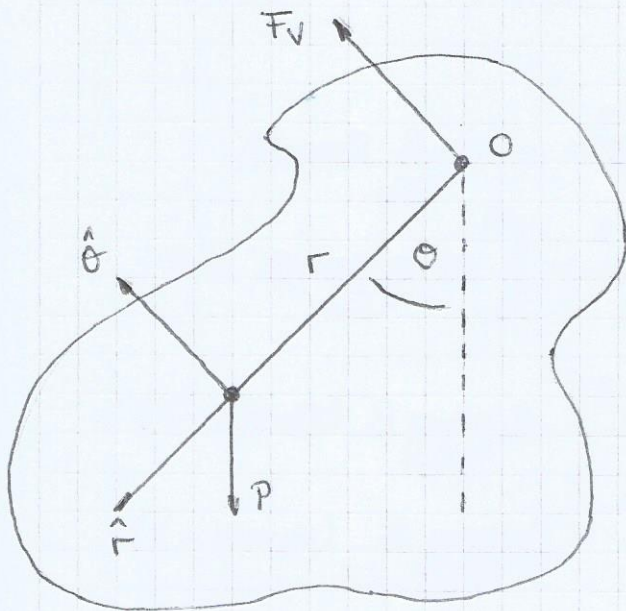
$$\frac{dL_0}{dt} = I_0 \Omega = I \Omega \dot{r}$$

Equilibrio: $\vec{F}^{\text{ext}} = 0$
 $M_o^{\text{ext}} = 0$

$$L_o = L_o' + \vec{r}_{oo'} \times \vec{p}$$

$$M_o^{\text{ext}} = M_o'^{\text{ext}} + \vec{r}_{oo'} \times \vec{F}^{\text{ext}}$$

Péndulo físico:



$$\vec{r} = R \hat{r}$$

$$\vec{F}_v + \vec{P} = m \vec{a}_{cm}$$

$$M_o^e = \vec{r} \times \vec{p} = R \hat{r} \times p (\cos\theta \hat{r} - \sin\theta \hat{\theta})$$

$$M_o^e = -m r g \sin\theta \hat{z}$$

$$M_o^e = I_o \gamma$$

$$I_o \gamma = -R m g \sin\theta \hat{z}$$

$$\gamma = \frac{-R m g \sin\theta}{I_o} \hat{z}$$

$$\ddot{\theta} + \frac{R m g \sin\theta}{I_o} = 0$$

Con $\theta < 1^\circ$ $\sin\theta = \theta$

Ecuación de movimiento:

$$\ddot{\theta} + \frac{R m g \theta}{I_o} = 0$$

$$T = \frac{2\pi}{\omega}$$

$$T = 2\pi \sqrt{\frac{I_o}{m g R}}$$

Por Steiner:

$$T = 2\pi \sqrt{\frac{I_{cm} + m R^2}{m g R}}$$

$$\vec{v}_{cm} = \vec{v}_0 + \vec{\Omega} \times (\vec{r}_{cm} - \vec{r}_0)$$

$$\vec{\Omega} = \omega \hat{z}$$

$$\vec{v}_{cm} = \vec{\Omega} \times \vec{R} = \omega R \hat{\theta}$$

$$\vec{r}_{cm} - \vec{r}_0 = R \hat{r}$$

$$\vec{a}_{cm} = \frac{d\vec{v}_{cm}}{dt} = \dot{\omega} R \hat{\theta} - \omega R \dot{\theta} \hat{r}$$

$$\vec{a}_{cm} = \dot{\theta} R \hat{\theta} - R \omega^2 \hat{r}$$

$$\hat{r}) \quad T_v + mg \cos \theta = -m r \omega^2$$

$$\ddot{\theta} = -\frac{mgR \sin \theta}{I_0}$$

$$\hat{\theta}) \quad T_v + mg \sin \theta = m \dot{\theta} R$$

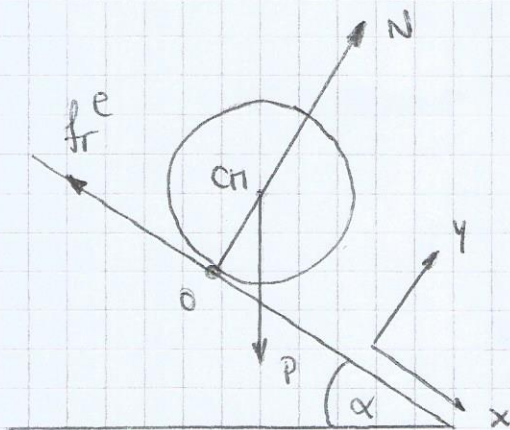
$$\frac{\dot{\theta}^2}{2} = \int_0^\theta \frac{mgR \sin \theta}{I_0} d\theta$$

$$T_v \hat{r} = -m r \omega^2 - mg \cos \theta$$

$$\dot{\theta}^2 = \frac{2mgR}{I_0} (\cos \theta - 1)$$

$$\omega^2 = \frac{2mgR}{I_0} (\cos \theta - 1)$$

Problema:



$$\vec{v}_{cm} = \omega \times (\vec{r}_{cm} - \vec{r}_0)$$

$\vec{O} \in E.I.R$

$$dW = \vec{f}_r^e \cdot d\vec{r}_0 = \vec{f}_r^e \cdot \vec{v}_0 dt = 0$$

$$\hat{x}) \quad mg \sin \alpha - f_r^e = m \vec{a}_{cm}$$

$$\hat{y}) \quad N - mg \cos \alpha = 0$$

$$M_0^{ext} = R \hat{y} \times mg (\sin \alpha \hat{x} - \cos \alpha \hat{y}) \quad I_0 \dot{\theta} = M_0^{ext}$$

$$M_0^{ext} = -mgR \sin \alpha \hat{z}$$

$$I_0 \dot{\theta} = -mgR \sin \alpha$$

$$\dot{\theta} = \frac{-mgR \sin \alpha}{I_0}$$

$$N_{cm} = \omega \hat{z} \times r \hat{y}$$

$$N_{cm} = -\omega R \hat{x}$$

$$\partial_{cm} = -\dot{\theta} R \hat{x}$$

$$\partial_{cm} = \frac{mg R^2 \sin \alpha}{I_0} \hat{x}$$

$$f_r^e = -\frac{m^2 g R^2}{I_0} \sin \alpha + mg \sin \alpha$$

$$f_r^e = mg \sin \alpha \left(\frac{-m R^2}{I_0} - 1 \right) = mg \sin \alpha \frac{I_{cm}}{I_0}$$

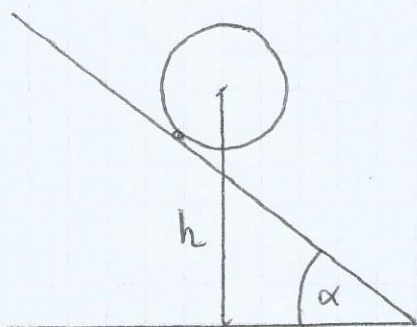
$$f_r^e \leq \mu N$$

$$f_r^e \leq \mu mg \cos \alpha$$

$$mg \sin \alpha \frac{I_{cm}}{I_0} \leq \mu mg \cos \alpha$$

$$\frac{\sin \alpha}{\cos \alpha} \frac{I_{cm}}{I_0} \leq \mu \frac{\cos \alpha}{\cos \alpha}$$

$$\tan \alpha \leq \mu \frac{I_0}{I_{cm}}$$



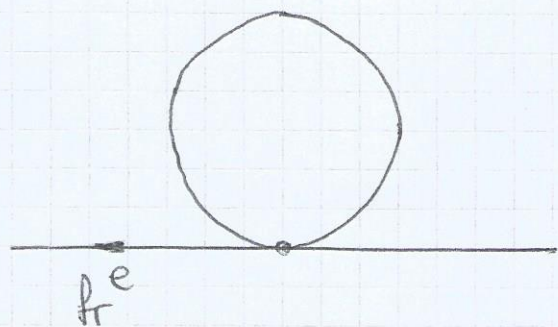
$$f_r^e$$

$$dW = f_r^e dr_o = f_r^e v_o dt = 0$$

$$E_{cm} = cte$$

$$E = \frac{1}{2} m v_{cm}^2 + \frac{1}{2} I_{cm} \Omega^2 + mgh$$

$$E = \frac{1}{2} m v_o^2 + \frac{1}{2} I_o \Omega^2 + mgh = cte \quad |v_{cm}| = \Omega R$$



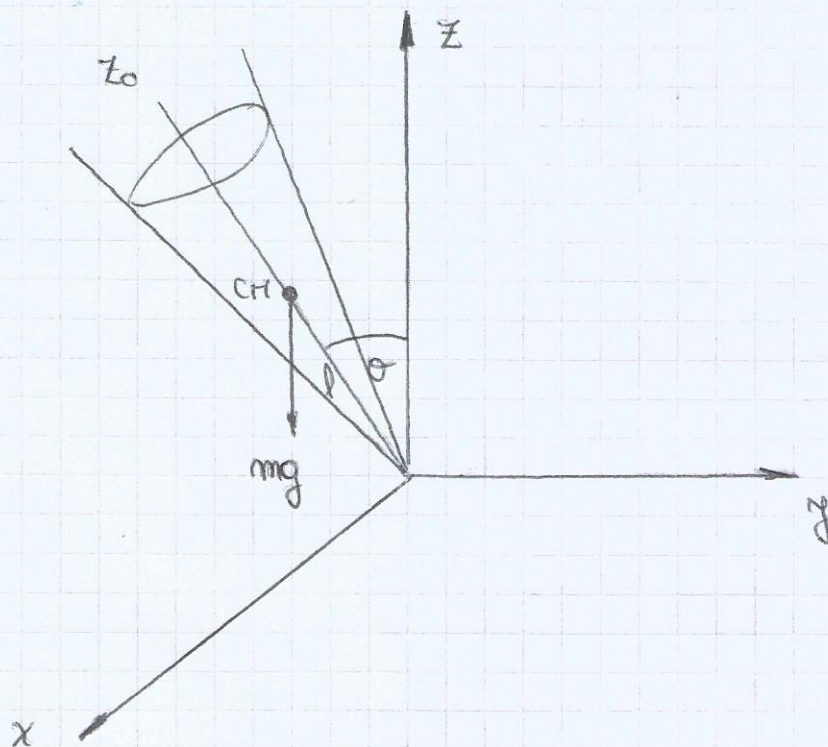
$$E_m = \frac{1}{2} I_0 \Omega^2 = \text{cte}$$

$$\Rightarrow \Omega^2 = \text{cte}$$

$$\Rightarrow \dot{\Omega} = 0$$

$$\Rightarrow \dot{r}^e = 0$$

Giroscopio:



$$L_0 = L_{cm} + R_{cm} \times p$$

$$L_{cm} = I_{cm} \omega$$

$$L_0 = I_{cm} \omega + R_{cm} \times p$$

Supongo que ω es muy grande (desprecio $R_{cm} \times p$)

$$L_0 \approx L_{cm}$$

$$dL = M_0^{\text{ext}} dt$$

$$dL = m g l \sin \theta dt \hat{n}$$

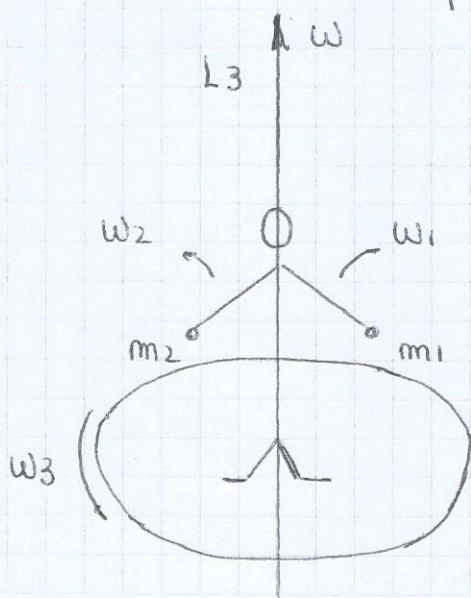
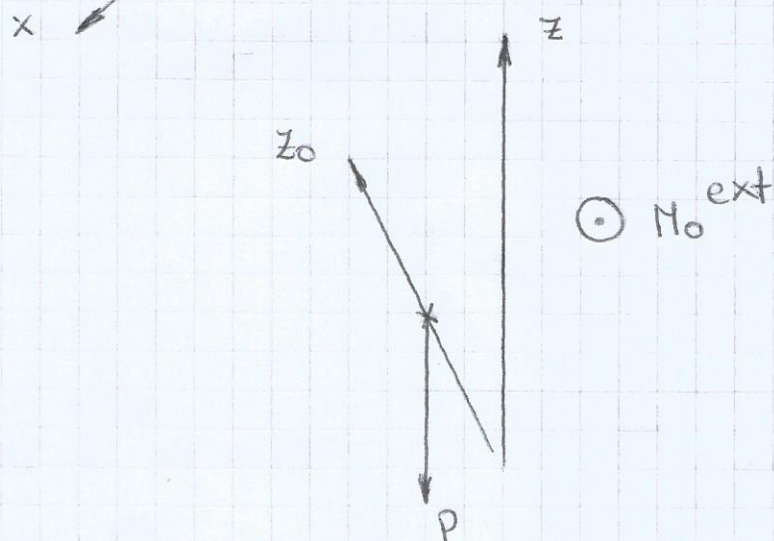
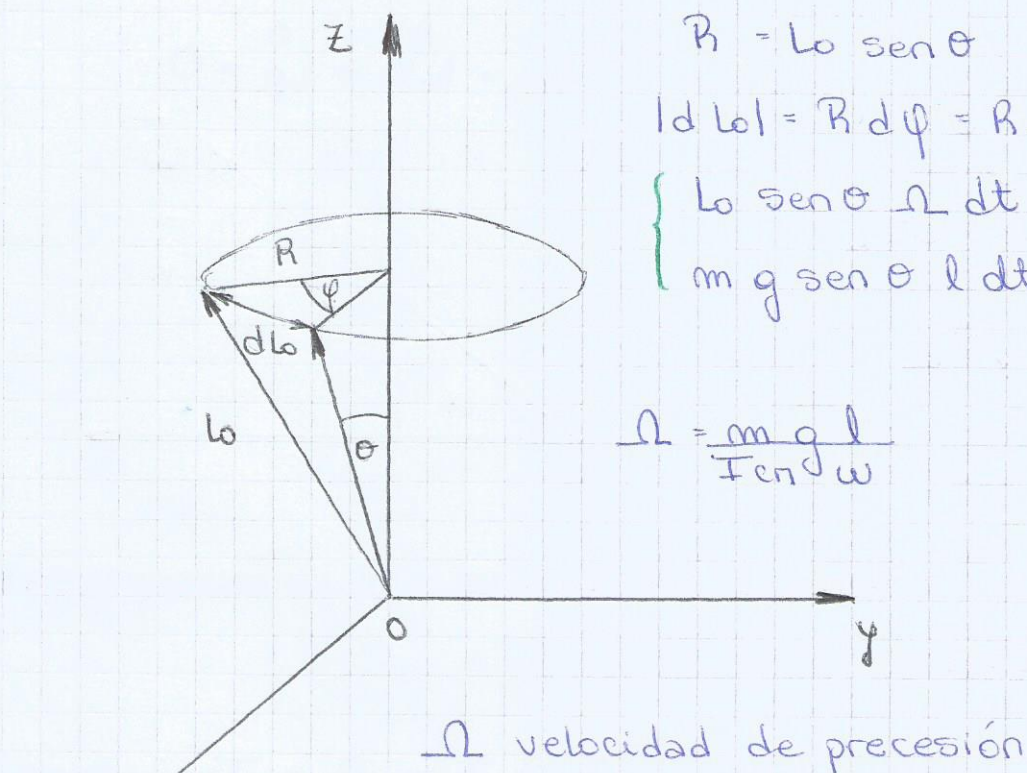
$$M_0^{\text{ext}} \perp L_{cm}$$

$$M_0 = R \times m g$$

$$M_0 = -l \hat{z}_0 \times m g \hat{z}$$

$$M_0 = m g l \sin \theta \hat{n}$$

($\hat{n}$ versor normal al plano que define z y z_0)



$L_{cm} = I_{cm} \omega = cte$
 "I_{cm} aumenta alejando las masas del eje de rotación"

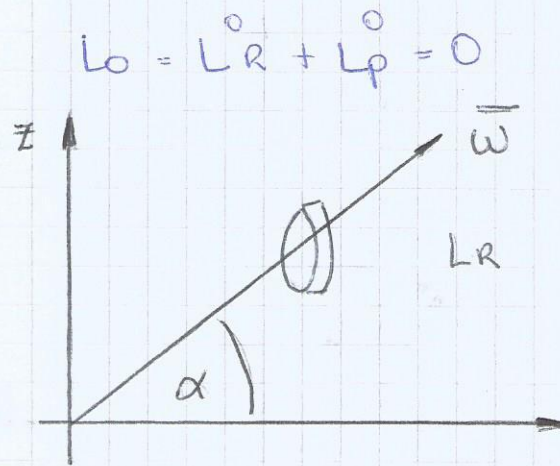
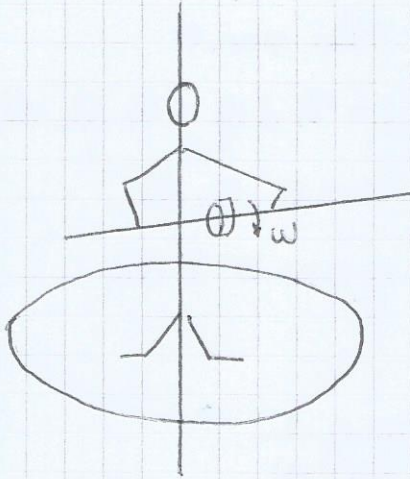
$$L_{cm} = L_{b1} + L_{b2} + L_h$$

$$L_{b1} = I_{b1} \omega_1$$

$$L_{b2} = I_{b2} \omega_2$$

$$L_{b3} = I_{b3} \omega_3$$

$$I_h \omega_3 = -I_{b1} \omega_1 - I_{b2} \omega_2$$



$$L_0 = L_R^0 + L_p^0 = 0$$

$$L_R^0 = L_e^{\text{cm}} + R_{\text{cm}} \times p \approx I_{\text{cm}} \omega \quad \text{Si } \omega \text{ est très grande}$$

$$L_{R\text{cm}} = I_{\text{cm}} \omega$$

$$L_0 = L_{R0} + L_{p0} = 0$$

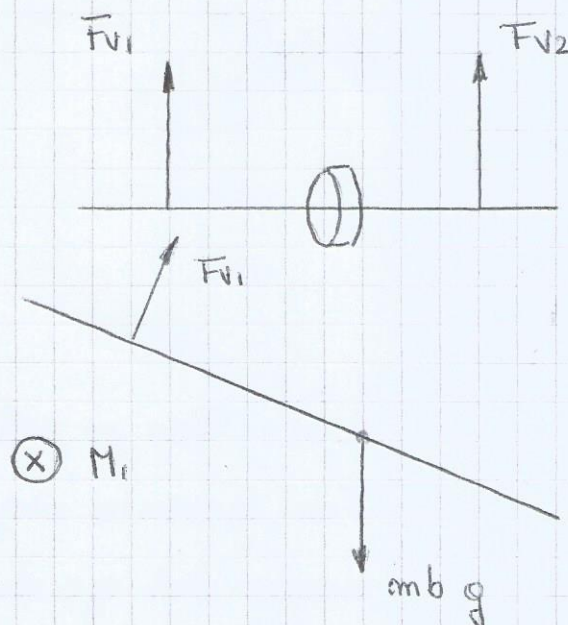
$$L_{pz} = I_{pz} \Omega = -I_{\text{cm}} \omega \sin \alpha \hat{z}$$

$$L_0 = I_{\text{cm}} \omega + L_{p0} = 0$$

$$\Omega = \frac{I_{\text{cm}} \omega \sin \alpha}{I_{pz}}$$

$$L_{p0} = -I_{\text{cm}} \omega$$

$$\text{Con } \alpha = 0$$



$$F_v \Rightarrow M_i \perp L_R$$

$$dL_R = M_i dt \perp L_R$$

$$\otimes M_i$$

$$mbg$$